

# 9D Distributed Maximum Likelihood Kalman Filter

## I. SYSTEM STATE AND KINEMATICS

Consider a cooperative localization system consisting of  $I$  mobile nodes indexed by  $i \in \{1, \dots, I\}$ . At discrete time step  $t$ , the state of node  $i$  is defined on the manifold  $\mathcal{M} = \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3)$  as:

$$\mathbf{x}_t^i = \begin{bmatrix} \mathbf{p}_t^i \\ \mathbf{v}_t^i \\ \mathbf{R}_t^i \end{bmatrix} \in \mathcal{M} \quad (1)$$

where  $\mathbf{p}_t^i \in \mathbb{R}^3$  denotes position,  $\mathbf{v}_t^i \in \mathbb{R}^3$  denotes linear velocity in the navigation frame ( $n$ -frame), and  $\mathbf{R}_t^i \in SO(3)$  is the rotation matrix transforming coordinates from the body frame ( $b$ -frame) to the  $n$ -frame.

The IMU measurements of vehicle  $i$  consist of specific force  $\tilde{\mathbf{a}}_t^i \in \mathbb{R}^3$  and angular velocity  $\tilde{\boldsymbol{\omega}}_t^i \in \mathbb{R}^3$  resolved in the  $b$ -frame:

$$\tilde{\mathbf{a}}_t^i = (\mathbf{R}_t^i)^\top (\mathbf{a}_t^i - \mathbf{g}) + \bar{\mathbf{a}}^i + \mathbf{e}_{a,t}^i \quad (2)$$

$$\tilde{\boldsymbol{\omega}}_t^i = \boldsymbol{\omega}_t^i + \bar{\boldsymbol{\omega}}^i + \mathbf{e}_{\omega,t}^i \quad (3)$$

where  $\mathbf{a}_t^i$  is the true acceleration in the  $n$ -frame,  $\mathbf{g}$  is the gravity vector,  $\bar{\mathbf{a}}^i, \bar{\boldsymbol{\omega}}^i$  are known constant sensor biases, and the measurement noises are Gaussian white sequences:

$$\mathbf{e}_{a,t}^i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_a^i), \quad \mathbf{e}_{\omega,t}^i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\omega^i) \quad (4)$$

The bias-corrected IMU signals driving the kinematic propagation are defined as:

$$\mathbf{a}_{m,t}^i \triangleq \tilde{\mathbf{a}}_t^i - \bar{\mathbf{a}}^i, \quad \boldsymbol{\omega}_{m,t}^i \triangleq \tilde{\boldsymbol{\omega}}_t^i - \bar{\boldsymbol{\omega}}^i \quad (5)$$

The Lie algebra  $\mathfrak{so}(3)$  is associated with  $3 \times 3$  skew-symmetric matrices via  $[\cdot]_\times : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ . The exponential map  $\exp : \mathbb{R}^3 \rightarrow SO(3)$  is evaluated via Rodrigues' formula:

$$\exp([\phi]_\times) = \mathbf{I}_3 + \frac{\sin \theta}{\theta} [\phi]_\times + \frac{1 - \cos \theta}{\theta^2} [\phi]_\times^2 \quad (6)$$

where  $\theta = \|\phi\|$ .

## II. NOMINAL AND ERROR STATE PROPAGATION

### A. Nominal State Propagation

Given the posterior state estimate  $\hat{\mathbf{x}}_{t|t}^i$  at time step  $t$  and sampling period  $\tau$ , the prior nominal state  $\hat{\mathbf{x}}_{t+1|t}^i$  propagates via numerical integration:

$$\hat{\mathbf{R}}_{t+1|t}^i = \hat{\mathbf{R}}_{t|t}^i \exp(\tau[\boldsymbol{\omega}_{m,t}^i]_\times) \quad (7)$$

$$\hat{\mathbf{v}}_{t+1|t}^i = \hat{\mathbf{v}}_{t|t}^i + \tau \left( \hat{\mathbf{R}}_{t|t}^i \mathbf{a}_{m,t}^i + \mathbf{g} \right) \quad (8)$$

$$\hat{\mathbf{p}}_{t+1|t}^i = \hat{\mathbf{p}}_{t|t}^i + \tau \hat{\mathbf{v}}_{t|t}^i + \frac{\tau^2}{2} \left( \hat{\mathbf{R}}_{t|t}^i \mathbf{a}_{m,t}^i + \mathbf{g} \right) \quad (9)$$

### B. Error State Dynamics and Covariance Propagation

Let the true state be parameterized by the error state components  $\delta \mathbf{p}_t^i$ ,  $\delta \mathbf{v}_t^i$ , and  $\delta \phi_t^i$ :

$$\mathbf{R}_t^i = \hat{\mathbf{R}}_{t|t}^i \exp([\delta \phi_{t|t}^i]_\times) \quad (10)$$

$$\mathbf{v}_t^i = \hat{\mathbf{v}}_{t|t}^i + \delta \mathbf{v}_{t|t}^i, \quad \mathbf{p}_t^i = \hat{\mathbf{p}}_{t|t}^i + \delta \mathbf{p}_{t|t}^i \quad (11)$$

The discrete-time linear error dynamics of node  $i$  are:

$$\boldsymbol{\theta}_{t+1|t}^i = \mathbf{A}_t^i \boldsymbol{\theta}_{t|t}^i + \mathbf{W}_t^i \mathbf{w}_t^i \quad (12)$$

where  $\boldsymbol{\theta}_t^i = [(\delta \mathbf{p}_t^i)^\top, (\delta \mathbf{v}_t^i)^\top, (\delta \phi_t^i)^\top]^\top \in \mathbb{R}^9$ ,  $\mathbf{w}_t^i = [(\mathbf{e}_{a,t}^i)^\top, (\mathbf{e}_{\omega,t}^i)^\top]^\top \in \mathbb{R}^6$ .

The state transition matrix  $\mathbf{A}_t^i \in \mathbb{R}^{9 \times 9}$  and noise input matrix  $\mathbf{W}_t^i \in \mathbb{R}^{9 \times 6}$  are defined compactly to fit the double-column width:

$$\mathbf{A}_t^i = \begin{bmatrix} \mathbf{I}_3 & \tau \mathbf{I}_3 & -\frac{\tau^2}{2} \hat{\mathbf{R}}_{t|t}^i [\mathbf{a}_{m,t}^i]_\times \\ \mathbf{0}_3 & \mathbf{I}_3 & -\tau \hat{\mathbf{R}}_{t|t}^i [\mathbf{a}_{m,t}^i]_\times \\ \mathbf{0}_3 & \mathbf{0}_3 & \exp(-\tau[\boldsymbol{\omega}_{m,t}^i]_\times) \end{bmatrix} \quad (13)$$

$$\mathbf{W}_t^i = \begin{bmatrix} -\frac{\tau^2}{2} \hat{\mathbf{R}}_{t|t}^i & \mathbf{0}_3 \\ -\tau \hat{\mathbf{R}}_{t|t}^i & \mathbf{0}_3 \\ \mathbf{0}_3 & -\tau \mathbf{J}_r(\tau \boldsymbol{\omega}_{m,t}^i) \end{bmatrix} \quad (14)$$

where the right Jacobian of  $SO(3)$  is resolved via:

$$\mathbf{J}_r(\phi) = \mathbf{I}_3 - \frac{1 - \cos \theta}{\theta^2} [\phi]_\times + \frac{\theta - \sin \theta}{\theta^3} [\phi]_\times^2 \quad (15)$$

The discrete local process noise covariance matrix  $\mathbf{Q}_t^i \in \mathbb{R}^{9 \times 9}$  and the error covariance propagation are:

$$\mathbf{Q}_t^i = \mathbf{W}_t^i \begin{bmatrix} \boldsymbol{\Sigma}_a^i & \mathbf{0}_3 \\ \mathbf{0}_3 & \boldsymbol{\Sigma}_\omega^i \end{bmatrix} (\mathbf{W}_t^i)^\top \quad (16)$$

$$\boldsymbol{\Sigma}_{t+1|t}^i = \mathbf{A}_t^i \boldsymbol{\Sigma}_{t|t}^i (\mathbf{A}_t^i)^\top + \mathbf{Q}_t^i \quad (17)$$

## III. DISTRIBUTED PRIOR COVARIANCE EXPANSION

Prior to optimization, node  $i$  receives the prior state  $\hat{\mathbf{p}}_{t|t-1}^j$  and prior covariance  $\boldsymbol{\Sigma}_{t|t-1}^j$  from each neighbor  $j \in \mathcal{N}_i$ . Assuming uncorrelated prior state errors across different targets, the joint neighborhood prior error covariance  $\bar{\boldsymbol{\Sigma}}_{t|t-1}^i$  and joint prior information matrix  $\bar{\boldsymbol{\Gamma}}_{t|t-1}^i$  are expanded block-diagonally as:

$$\bar{\boldsymbol{\Sigma}}_{t|t-1}^i = \text{blockdiag} \left( \boldsymbol{\Sigma}_{t|t-1}^i, \{\boldsymbol{\Sigma}_{t|t-1}^j\}_{j \in \mathcal{N}_i} \right) \quad (18)$$

$$\begin{aligned} \bar{\boldsymbol{\Gamma}}_{t|t-1}^i &= \left( \bar{\boldsymbol{\Sigma}}_{t|t-1}^i \right)^{-1} \\ &= \text{blockdiag} \left( \boldsymbol{\Gamma}_{t|t-1}^i, \{\boldsymbol{\Gamma}_{t|t-1}^j\}_{j \in \mathcal{N}_i} \right) \end{aligned} \quad (19)$$

where  $\boldsymbol{\Gamma}_{t|t-1}^i = (\boldsymbol{\Sigma}_{t|t-1}^i)^{-1}$  and  $\boldsymbol{\Gamma}_{t|t-1}^j = (\boldsymbol{\Sigma}_{t|t-1}^j)^{-1}$ . Because the nominal state has zero prior error mean, the joint prior information vector of the neighborhood error state is defined as:

$$\bar{\boldsymbol{\gamma}}_{t|t-1}^i = \mathbf{0} \quad (20)$$

#### IV. DISTRIBUTED GAUSS-NEWTON OPTIMIZATION

##### A. Problem Formulation and Variable Definition

Let  $\mathcal{N}_i$  denote the set of spatial neighbors of node  $i$ . The set of nodes whose state errors are estimated by node  $i$  is defined as the local index set:

$$\mathcal{U}_i = \{i\} \cup \mathcal{N}_i \quad (21)$$

At consensus iteration  $k$ , the local state error vector maintained at node  $i$  is stacked as:

$$\mathbf{s}_k^i = \text{col} \left( \mathbf{s}_k^{i,c} \right)_{c \in \mathcal{U}_i} \in \mathbb{R}^{3|\mathcal{U}_i|} \quad (22)$$

where  $\mathbf{s}_k^{i,c} \in \mathbb{R}^3$  represents node  $i$ 's estimate of the 3D position error  $\delta \mathbf{p}^c$  of node  $c$  at iteration  $k$ .

The local objective function  $F^i(\mathbf{s}_k^i)$  is formulated using absolute anchor measurements and relative inter-node measurements:

$$F^i(\mathbf{s}_k^i) = J_{\text{anc}}^i(\mathbf{s}_k^i) + J_{\text{inter}}^i(\mathbf{s}_k^i) \quad (23)$$

where the absolute anchor ranging cost is:

$$J_{\text{anc}}^i(\mathbf{s}_k^i) = \frac{1}{2\sigma_s^2} \sum_{l=1}^K \left( l_t^l - \left\| \hat{\mathbf{p}}_{t|t-1}^i + \mathbf{s}_k^{i,i} - \mathbf{c}^l \right\| \right)^2 \quad (24)$$

and the relative inter-node ranging cost is:

$$J_{\text{inter}}^i(\mathbf{s}_k^i) = \frac{1}{2\sigma_z^2} \sum_{j \in \mathcal{N}_i} \left( z_t^{i,j} - \left\| (\hat{\mathbf{p}}_{t|t-1}^i + \mathbf{s}_k^{i,i}) - (\hat{\mathbf{p}}_{t|t-1}^j + \mathbf{s}_k^{j,j}) \right\| \right)^2 \quad (25)$$

For each physical variable  $\delta \mathbf{p}^c$ , the consensus sub-network consisting of all nodes that maintain an estimate of  $\delta \mathbf{p}^c$  is defined as:

$$\mathcal{U}_c = \{c\} \cup \mathcal{N}_c \quad (26)$$

##### B. Initialization

At iteration  $k = 1$ , the local variables are initialized as:

$$\mathbf{s}_1^{i,c} = \mathbf{0}, \quad \forall c \in \mathcal{U}_i \quad (27)$$

$$\mathbf{g}_1^{i,c} = \nabla_c F^i(\mathbf{s}_1^i), \quad \forall c \in \mathcal{U}_i \quad (28)$$

$$\mathbf{H}_1^{i,(c,d)} = \nabla_{c,d}^2 F^i(\mathbf{s}_1^i), \quad \forall c, d \in \mathcal{U}_i \quad (29)$$

where  $\nabla_c F^i$  and  $\nabla_{c,d}^2 F^i$  denote the first and second partial derivatives of the objective function  $F^i$  with respect to variables  $c$  and  $d$ .

##### C. Localized Weight Matrices

For each variable  $c \in \{1, \dots, I\}$ , the localized Metropolis-Hastings weight matrix  $\mathbf{W}_c = [w_c^{i,j}] \in \mathbb{R}^{|\mathcal{U}_c| \times |\mathcal{U}_c|}$  is configured as:

$$w_c^{i,j} = \begin{cases} \frac{1}{1+\max(d_c^i, d_c^j)} & \text{if } j \in \mathcal{U}_c \cap \mathcal{N}_i \\ 1 - \sum_{l \in \mathcal{U}_c \cap \mathcal{N}_i} \frac{1}{1+\max(d_c^i, d_c^l)} & \text{if } j = i \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

where  $d_c^i = |\mathcal{U}_c \cap \mathcal{U}_i| - 1$  represents the localized communication degree of node  $i$  within subnetwork  $\mathcal{U}_c$ .

Similarly, for each cross-variable pair  $(c, d)$  where  $c, d \in \mathcal{U}_i$ , the weight matrix  $\mathbf{W}_{c,d} = [w_{c,d}^{i,j}] \in \mathbb{R}^{|\mathcal{U}_{c,d}| \times |\mathcal{U}_{c,d}|}$  is constructed over the subnetwork  $\mathcal{U}_{c,d} = \mathcal{U}_c \cap \mathcal{U}_d$  as:

$$w_{c,d}^{i,j} = \begin{cases} \frac{1}{1+\max(d_{c,d}^i, d_{c,d}^j)} & \text{if } j \in \mathcal{U}_{c,d} \cap \mathcal{N}_i \\ 1 - \sum_{l \in \mathcal{U}_{c,d} \cap \mathcal{N}_i} \frac{1}{1+\max(d_{c,d}^i, d_{c,d}^l)} & \text{if } j = i \\ 0 & \text{otherwise} \end{cases} \quad (31)$$

where  $d_{c,d}^i = |\mathcal{U}_{c,d} \cap \mathcal{U}_i| - 1$  is the local communication degree of node  $i$  within subnetwork  $\mathcal{U}_{c,d}$ .

##### D. Consensus and Local Newton Iterations

For iterations  $k \geq 1$ , each node  $i$  updates its local error state, gradient tracking, and Hessian tracking:

1) *State Consensus Update*: For each variable  $c \in \mathcal{U}_i$ :

$$\mathbf{s}_{k+1}^{i,c} = \sum_{j \in \mathcal{U}_c \cap \mathcal{U}_i} w_{c,d}^{i,j} \mathbf{s}_k^{j,c} - \alpha_k^i [B(\mathbf{H}_k^i)^{-1} \mathbf{g}_k^i]_c \quad (32)$$

where  $\alpha_k^i$  is the step-size,  $\mathbf{g}_k^i = \text{col}(\mathbf{g}_k^{i,c})_{c \in \mathcal{U}_i}$ , and  $\mathbf{H}_k^i$  is the local block diagonal-modified Hessian constructed from blocks  $\mathbf{H}_k^{i,(c,d)}$  for  $c, d \in \mathcal{U}_i$ .  $B(\cdot)$  is the positive definiteness projection map.

2) *Dynamic Gradient Consensus Tracking*: For each variable  $c \in \mathcal{U}_i$ :

$$\mathbf{g}_{k+1}^{i,c} = \sum_{j \in \mathcal{U}_c \cap \mathcal{U}_i} w_{c,d}^{i,j} \left[ \mathbf{g}_k^{j,c} + \nabla_c F^j(\mathbf{s}_{k+1}^j) - \nabla_c F^j(\mathbf{s}_k^j) \right] \quad (33)$$

3) *Dynamic Hessian Consensus Tracking*: For each pair  $(c, d)$  where  $c, d \in \mathcal{U}_i$  and  $\mathcal{U}_{c,d} \cap \mathcal{U}_i \neq \emptyset$ :

$$\mathbf{H}_{k+1}^{i,(c,d)} = \sum_{j \in \mathcal{U}_{c,d} \cap \mathcal{U}_i} w_{c,d}^{i,j} \left[ \mathbf{H}_k^{j,(c,d)} + \nabla_{c,d}^2 F^j(\mathbf{s}_{k+1}^j) - \nabla_{c,d}^2 F^j(\mathbf{s}_k^j) \right] \quad (34)$$

##### E. Information Exchange Packets

At each iteration  $k$ , node  $i$  transmits the message packet  $\mathcal{M}_k^{i \rightarrow j}$  to out-neighbor  $j \in \mathcal{N}_i$ :

$$\mathcal{M}_k^{i \rightarrow j} = \left\{ \mathcal{S}_k^{i \rightarrow j}, \mathcal{G}_k^{i \rightarrow j}, \mathcal{H}_k^{i \rightarrow j} \right\} \quad (35)$$

where the packet blocks are formulated as:

$$\mathcal{S}_k^{i \rightarrow j} = \left\{ \mathbf{s}_k^{i,c} \mid c \in \mathcal{U}_i \cap \mathcal{U}_j \right\} \quad (36)$$

$$\mathcal{G}_k^{i \rightarrow j} = \left\{ \left( \mathbf{g}_k^{i,c}, \nabla_c F^i(\mathbf{s}_{k+1}^i) - \nabla_c F^i(\mathbf{s}_k^i) \right) \mid c \in \mathcal{U}_i \cap \mathcal{U}_j \right\} \quad (37)$$

$$\mathcal{H}_k^{i \rightarrow j} = \left\{ \left( \mathbf{H}_k^{i,(c,d)}, \nabla_{c,d}^2 F^i(\mathbf{s}_{k+1}^i) - \nabla_{c,d}^2 F^i(\mathbf{s}_k^i) \right) \mid c, d \in \mathcal{U}_i \cap \mathcal{U}_j \right\} \quad (38)$$

The iteration terminates when  $\|\mathbf{s}_{k+1}^i - \mathbf{s}_k^i\| < \epsilon$ , yielding the converged error state estimate  $\mathbf{s}_{DN}^i = \mathbf{s}_{k+1}^i$ .

## V. POSTERIOR INFORMATION FUSION AND MARGINALIZATION

### A. Likelihood Information Extraction and State Space Lifting

Upon convergence of the Distributed Gauss-Newton (D-GN) iterations, the local consensus Hessian blocks  $\mathbf{H}_{DN}^{i,(c,d)}$  represent the average measurement curvature over the subnetworks. To reconstruct the absolute total Fisher information, the joint neighborhood likelihood information matrix  $\bar{\mathbf{\Lambda}}_t^i \in \mathbb{R}^{3|\mathcal{U}_i| \times 3|\mathcal{U}_i|}$  and vector  $\bar{\boldsymbol{\lambda}}_t^i \in \mathbb{R}^{3|\mathcal{U}_i|}$  are extracted by rescaling with the subnetwork sizes:

$$\bar{\mathbf{\Lambda}}_t^{i,(c,d)} = |\mathcal{U}_{c,d}| \cdot \mathbf{H}_{DN}^{i,(c,d)}, \quad \forall c, d \in \mathcal{U}_i \quad (39)$$

$$\bar{\mathbf{\Lambda}}_t^i = \left[ \bar{\mathbf{\Lambda}}_t^{i,(c,d)} \right]_{c,d \in \mathcal{U}_i} \quad (40)$$

$$\bar{\boldsymbol{\lambda}}_t^i = \bar{\mathbf{\Lambda}}_t^i \mathbf{s}_{DN}^i \quad (41)$$

where  $\mathbf{s}_{DN}^i$  is the scale-free converged local state error.

Since the prior information matrix  $\bar{\mathbf{\Gamma}}_{t|t-1}^i$  is defined over the full 9D error state of each node, a tangent space selection matrix  $\bar{\boldsymbol{\pi}}^i \in \mathbb{R}^{3|\mathcal{U}_i| \times 9|\mathcal{U}_i|}$  is constructed to lift the 3D position likelihood into the 9D joint error space:

$$\bar{\boldsymbol{\pi}}^i = \text{blockdiag}(\boldsymbol{\pi}^c)_{c \in \mathcal{U}_i} \quad (42)$$

where each local block  $\boldsymbol{\pi}^c \in \mathbb{R}^{3 \times 9}$  isolates position error:

$$\boldsymbol{\pi}^c = [\mathbf{I}_3 \quad \mathbf{0}_3 \quad \mathbf{0}_3] \quad (43)$$

The lifted joint neighborhood likelihood matrix  $\bar{\mathbf{\Lambda}}_{\text{lifted}}^i \in \mathbb{R}^{9|\mathcal{U}_i| \times 9|\mathcal{U}_i|}$  and vector  $\bar{\boldsymbol{\lambda}}_{\text{lifted}}^i \in \mathbb{R}^{9|\mathcal{U}_i|}$  are evaluated as:

$$\bar{\mathbf{\Lambda}}_{\text{lifted}}^i = (\bar{\boldsymbol{\pi}}^i)^\top \bar{\mathbf{\Lambda}}_t^i \bar{\boldsymbol{\pi}}^i \quad (44)$$

$$\bar{\boldsymbol{\lambda}}_{\text{lifted}}^i = (\bar{\boldsymbol{\pi}}^i)^\top \bar{\boldsymbol{\lambda}}_t^i \quad (45)$$

### B. Posterior Information Fusion

The joint posterior information matrix  $\bar{\mathbf{\Gamma}}_{t|t}^i \in \mathbb{R}^{9|\mathcal{U}_i| \times 9|\mathcal{U}_i|}$  and information vector  $\bar{\boldsymbol{\gamma}}_{t|t}^i \in \mathbb{R}^{9|\mathcal{U}_i|}$  of the neighborhood are fused additively:

$$\bar{\mathbf{\Gamma}}_{t|t}^i = \bar{\mathbf{\Gamma}}_{t|t-1}^i + \bar{\mathbf{\Lambda}}_{\text{lifted}}^i \quad (46)$$

$$\bar{\boldsymbol{\gamma}}_{t|t}^i = \bar{\boldsymbol{\gamma}}_{t|t-1}^i + \bar{\boldsymbol{\lambda}}_{\text{lifted}}^i = \bar{\boldsymbol{\lambda}}_{\text{lifted}}^i \quad (47)$$

### C. Schur Complement Marginalization

Let the neighborhood post-fusion variables be partitioned into the ego state components (subscript  $i$ ) and neighbor components (subscript  $\mathcal{N}$ ):

$$\bar{\mathbf{\Gamma}}_{t|t}^i = \begin{bmatrix} \mathbf{\Gamma}_{ii} & \mathbf{\Gamma}_{i\mathcal{N}} \\ \mathbf{\Gamma}_{\mathcal{N}i} & \mathbf{\Gamma}_{\mathcal{N}\mathcal{N}} \end{bmatrix}, \quad \bar{\boldsymbol{\gamma}}_{t|t}^i = \begin{bmatrix} \boldsymbol{\gamma}_i \\ \boldsymbol{\gamma}_{\mathcal{N}} \end{bmatrix} \quad (48)$$

where  $\mathbf{\Gamma}_{ii} \in \mathbb{R}^{9 \times 9}$  represents the ego block, and  $\mathbf{\Gamma}_{\mathcal{N}\mathcal{N}} \in \mathbb{R}^{9|\mathcal{N}_i| \times 9|\mathcal{N}_i|}$  represents the neighbor blocks.

Applying the Schur complement to marginalize out the neighbors yields the posterior information matrix  $\mathbf{\Gamma}_{t|t}^i \in \mathbb{R}^{9 \times 9}$  and vector  $\boldsymbol{\gamma}_{t|t}^i \in \mathbb{R}^9$  of the ego node:

$$\mathbf{\Gamma}_{t|t}^i = \mathbf{\Gamma}_{ii} - \mathbf{\Gamma}_{i\mathcal{N}} \mathbf{\Gamma}_{\mathcal{N}\mathcal{N}}^{-1} \mathbf{\Gamma}_{\mathcal{N}i} \quad (49)$$

$$\boldsymbol{\gamma}_{t|t}^i = \boldsymbol{\gamma}_i - \mathbf{\Gamma}_{i\mathcal{N}} \mathbf{\Gamma}_{\mathcal{N}\mathcal{N}}^{-1} \boldsymbol{\gamma}_{\mathcal{N}} \quad (50)$$

### D. Posterior State Update and Retraction

The posterior state error covariance of the ego node is resolved as:

$$\boldsymbol{\Sigma}_{t|t}^i = \left( \mathbf{\Gamma}_{t|t}^i \right)^{-1} \quad (51)$$

The full 9D posterior error correction vector  $\boldsymbol{\theta}_{t|t}^i \in \mathbb{R}^9$  is extracted via:

$$\boldsymbol{\theta}_{t|t}^i = \boldsymbol{\Sigma}_{t|t}^i \boldsymbol{\gamma}_{t|t}^i \quad (52)$$

By parsing  $\boldsymbol{\theta}_{t|t}^i = [(\delta \mathbf{p}^i)^\top, (\delta \mathbf{v}^i)^\top, (\delta \boldsymbol{\phi}^i)^\top]^\top$ , the nominal prior state estimates are updated and mapped onto the manifold  $\mathcal{M}$  via the retraction operator:

$$\hat{\mathbf{p}}_{t|t}^i = \hat{\mathbf{p}}_{t|t-1}^i + \delta \mathbf{p}^i \quad (53)$$

$$\hat{\mathbf{v}}_{t|t}^i = \hat{\mathbf{v}}_{t|t-1}^i + \delta \mathbf{v}^i \quad (54)$$

$$\hat{\mathbf{R}}_{t|t}^i = \hat{\mathbf{R}}_{t|t-1}^i \exp([\delta \boldsymbol{\phi}^i]_\times) \quad (55)$$